the 1840s, humans had hunted the great auk to extinction to satisfy demand for its feathers, eggs, and meat.

Also susceptible to overharvesting are large organisms with low reproductive rates, such as elephants, whales, and rhinoceroses. The decline of Earth's largest terrestrial animals, the African elephants, is a classic example of the impact of overhunting. Largely because of the trade in ivory, elephant populations have been declining in most of Africa for the last 50 years. An international ban on the sale of new ivory resulted in increased poaching (illegal hunting), so the ban had little effect in much of central and eastern Africa. From 2006 to 2015, the number of elephants dropped by 110,000, a 22% decline (from 525,000 to 415,000). Only in South Africa, where once-decimated herds have been well protected for nearly a century, have elephant populations been stable or increasing (see Figure 53.8). If such protections are not extended to other regions, elephants could become extinct in the wild across most of Africa.

Conservation biologists are increasingly using the tools of molecular genetics to track the origins of tissues harvested from endangered species. For example, researchers have used DNA isolated from samples of elephant dung to construct a DNA reference map for the African elephant (*Loxodonta africana*). By comparing this reference map with DNA isolated from ivory harvested legally or by poachers, they can determine to within a few hundred kilometers where the elephants were killed **(Figure 56.8)**. Such work in Zambia suggested that poaching rates were 30 times higher than previously estimated, a finding that has stimulated improved anti-poaching efforts by the Zambian government. Similarly, biologists using phylogenetic analyses of mitochondrial DNA (mtDNA) showed that some whale meat sold in Japanese fish markets came from illegally harvested species, including fin and humpback whales, which are endangered (see Figure 26.6).

▼ Figure 56.8 Ecological forensics and elephant poaching. These severed tusks were part of an illegal shipment of ivory intercepted on its way from Africa to Singapore in 2002. DNA-based evidence showed that the thousands of elephants killed for the tusks came from a relatively narrow east-west band centered in Zambia rather than from across Africa.

MAKE CONNECTIONS *Figure 26.6 describes a similar example in which conservation biologists used DNA analyses to compare harvested samples of whale meat with a reference DNA database. How are these examples similar, and how are they different? What limitations might there be to using such forensic methods in other suspected cases of poaching?*

▼ Figure 56.9 Overharvesting. Atlantic bluefin tuna are auctioned in a Japanese fish market.

Many commercially important fish populations, once thought to be inexhaustible, have been decimated by overfishing. Demands from an increasing human population for protein-rich food, coupled with new harvesting technologies, such as long-line fishing and modern trawlers, have reduced these fish populations to levels that cannot sustain further exploitation. Until the past few decades, the Atlantic bluefin tuna was considered a sport fish of little commercial value—just a few cents per pound for use in cat food. In the 1980s, however, wholesalers began flying fresh, iced bluefin to Japan for sushi and sashimi. In that market, the fish can now bring over $500 per pound **(Figure 56.9)**. With increased harvesting spurred by such high prices, it took just ten years to reduce the western Atlantic bluefin population to less than 20% of its 1980 size.

Global Change

The fourth threat to biodiversity, global change, alters the fabric of Earth's ecosystems at regional to global scales. Global change includes alterations in climate, atmospheric chemistry, and broad ecological systems that reduce the capacity of Earth to sustain life.

One of the first types of global change to cause concern was *acid precipitation*, which is rain, snow, or fog with a pH less than 5.2. The burning of wood and fossil fuels releases oxides of sulfur and nitrogen that react with water in air, forming sulfuric and nitric acids. The acids eventually fall to Earth's surface, where they cause chemical reactions that decrease nutrient supplies and increase concentrations of toxic metals. These changes to the soil and water harm some terrestrial and aquatic organisms.

In the 1960s, ecologists determined that lake-dwelling organisms in eastern Canada were dying because of acid precipitation caused by air pollution from factories in the Midwestern United States. Newly hatched lake trout, for instance, die when the pH drops below 5.4. Similarly, lakes and streams in southern Norway and Sweden were losing fish